

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-145367147})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}243 + \mathbf{Z}(146 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 36341787 = 0$, of discriminant -145367147 . Class group invariants $(125 \quad 5 \quad 5)$, class number 3125; the three stored generator ideals (HNF) are $\begin{pmatrix} 11 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1461)y^3 + (40s - 22841)y^2 + (-496s + 1318012)y + (2091s - 10470351)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2919y^8 - 39796y^7 + 41204581y^6 - 2866846824y^5 \\ & + 90908762347y^4 - 1623567054940y^3 + 17234257097955y^2 - 1029748 \\ & 98471300y + 268502861365407 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right)s, \quad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $148071703930416 = 2^4 \cdot 3^4 \cdot 7^2 \cdot 2331690979$:

$$\begin{pmatrix} 1762758380124 & 295958620524 & 698920297134 & 1022332005717 & 1706469471437 & 857437419920 & 173977642215 & 339849787429 & 494138011360 & 983301729799 \\ 0 & 28 & 16 & 17 & 13 & 17 & 25 & 8 & 14 & 10 \\ 0 & 0 & 3 & 2 & 2 & 1 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 226 factors with signed exponents of absolute value up to 1842929065801238; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1461)y^3 + (40s - 22841)y^2 + (-496s + 1318012)y + (2091s - 10470351)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2919y^8 - 39796y^7 + 41204581y^6 - 2866846824y^5 \\ & + 90908762347y^4 - 1623567054940y^3 + 17234257097955y^2 - 1029748 \\ & 98471300y + 268502861365407 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right)s, \quad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $323079881298417 = 3^7 \cdot 37 \cdot 3992633143$:

$$\begin{pmatrix} 1329546836619 & 488313146556 & 763379049792 & 995756518875 & 275476837032 & 764482833148 & 482091980565 & 150952630306 & 418172374833 & 324639002923 \\ 0 & 9 & 0 & 1 & 0 & 6 & 8 & 8 & 0 & 8 \\ 0 & 0 & 3 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 2 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 211 factors with signed exponents of absolute value up to 2568514874133254; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1461)y^3 + (40s - 22841)y^2 + (-496s + 1318012)y + (2091s - 10470351)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 2919y^8 - 39796y^7 + 41204581y^6 - 2866846824y^5 \\ + 90908762347y^4 - 1623567054940y^3 + 17234257097955y^2 - 1029748 \\ 98471300y + 268502861365407$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ -1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right)s, \quad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $49278904535232 = 2^6 \cdot 3^7 \cdot 37 \cdot 73 \cdot 130349$:

$$\begin{pmatrix} 114071538276 & 14127364152 & 54932617404 & 50709401370 & 32269400549 & 355607676 & 84245706094 & 40054680368 & 12560127955 & 2119595412 \\ 0 & 12 & 0 & 6 & 9 & 3 & 8 & 7 & 3 & 4 \\ 0 & 0 & 6 & 4 & 2 & 3 & 3 & 3 & 5 & 3 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 2 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 234 factors with signed exponents of absolute value up to 2131474240736027; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (s - 3032)y^3 + (36s - 80152)y^2 + (341s + 1613825)y + (-378s + 9102838)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 6063y^8 - 160268y^7 + 48759770y^6 + 3120666386y^5 + 68519 \\ 857431y^4 + 550927965822y^3 + 4382899360977y^2 + 20014501791156y \\ + 88050878675188$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right) s, \quad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $57625467275575 = 5^2 \cdot 895691 \cdot 2573453$:

$$\begin{pmatrix} 11525093455115 & 9445693474315 & 6933200577284 & 6244744795605 & 11442978292550 & 2260349059294 & 9434205862078 & 773797320163 & 629768752977 & 7962598032971 \\ 0 & 5 & 0 & 4 & 3 & 0 & 4 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 316 factors with signed exponents of absolute value up to 20435587795; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (s - 3032)y^3 + (36s - 80152)y^2 + (341s + 1613825)y + (-378s + 9102838)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6063y^8 - 160268y^7 + 48759770y^6 + 3120666386y^5 + 68519 \\ & 857431y^4 + 550927965822y^3 + 4382899360977y^2 + 20014501791156y \\ & + 88050878675188 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right) s, \quad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $57625467275575 = 5^2 \cdot 895691 \cdot 2573453$:

$$\begin{pmatrix} 11525093455115 & 9445693474315 & 6933200577284 & 6244744795605 & 11442978292550 & 2260349059294 & 9434205862078 & 773797320163 & 629768752977 & 7962598032971 \\ 0 & 5 & 0 & 4 & 3 & 0 & 4 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 309 factors with signed exponents of absolute value up to 32788251594; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (4 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + (s - 3032)y^3 + (36s - 80152)y^2 + (341s + 1613825)y + (-378s + 9102838)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 6063y^8 - 160268y^7 + 48759770y^6 + 3120666386y^5 + 68519 \\ & 857431y^4 + 550927965822y^3 + 4382899360977y^2 + 20014501791156y \\ & + 88050878675188 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 0 \ 0 \ 0 \ -1 \ 1 \ 1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right)s, \quad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $139620653811925 = 5^2 \cdot 11^2 \cdot 13 \cdot 1009 \cdot 1297 \cdot 2713$:

$$\begin{pmatrix} 2538557342035 & 2193343278236 & 2412385824361 & 1348386902375 & 2324460976244 & 907558887807 & 1556954595250 & 1019117867398 & 2419996922939 & 2499913022996 \\ 0 & 11 & 6 & 10 & 9 & 5 & 7 & 6 & 2 & 7 \\ 0 & 0 & 5 & 3 & 4 & 2 & 4 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 314 factors with signed exponents of absolute value up to 32997314158; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 6193y^3 + (-12s + 6)y^2 + 11953708y + (-17232s + 8616)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} + 12386y^8 + 62260665y^6 + 153291844580y^4 + 157920935011888 \\ & \quad y^2 + 10791396444964032 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right)s, \quad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $733382257797396 = 2^2 \cdot 3^4 \cdot 29^2 \cdot 33829 \cdot 79561$:

$$\begin{pmatrix} 1404946854018 & 154977889524 & 940265332011 & 956854072224 & 719905291077 & 158133111898 & 865475666966 & 131303827082 & 1345682429978 & 398580538653 \\ 0 & 87 & 3 & 36 & 6 & 35 & 15 & 83 & 58 & 48 \\ 0 & 0 & 3 & 2 & 1 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 229 factors with signed exponents of absolute value up to 3626337429; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 6193y^3 + (-12s + 6)y^2 + 11953708y + (-17232s + 8616)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 12386y^8 + 62260665y^6 + 153291844580y^4 + 157920935011888y^2 + 10791396444964032$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right)s, \quad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18686585256372 = 2^2 \cdot 3^5 \cdot 11 \cdot 1747716541$:

$$\begin{pmatrix} 346047875118 & 334467155754 & 84740082138 & 72503129515 & 207783862989 & 204701488444 & 307652418027 & 215616585110 & 134219496680 & 175077894901 \\ 0 & 9 & 3 & 0 & 0 & 1 & 8 & 5 & 2 & 3 \\ 0 & 0 & 3 & 1 & 1 & 2 & 2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 224 factors with signed exponents of absolute value up to 2074955016; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 6193y^3 + (-12s + 6)y^2 + 11953708y + (-17232s + 8616)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 12386y^8 + 62260665y^6 + 153291844580y^4 + 157920935011888y^2 + 10791396444964032$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right)s, \quad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $33273839753748 = 2^2 \cdot 3^4 \cdot 11^2 \cdot 29 \cdot 29266753$:

$$\begin{pmatrix} 168049695726 & 107035048428 & 139837805217 & 154742764422 & 156952346499 & 127293816998 & 129365504829 & 140880265544 & 55121356777 & 102705190209 \\ 0 & 33 & 21 & 30 & 30 & 3 & 29 & 31 & 7 & 30 \\ 0 & 0 & 3 & 2 & 1 & 2 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 225 factors with signed exponents of absolute value up to 9842927219; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 47)y^3 + (-32s - 65038)y^2 + (569s + 347968)y + (-6358s - 6782218)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 89y^8 - 129922y^7 + 37300670y^6 - 2334788134y^5 + \\ & 82797719975y^4 - 1830224060950y^3 + 27558076586615y^2 - 26767357 \\ & 2155582y + 1515127960082636 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right)s, \quad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1125460241408 = 2^{10} \cdot 59 \cdot 18628513$:

$$\begin{pmatrix} 17585316272 & 419466988 & 5022817258 & 6603836502 & 15893624172 & 5209419996 & 12971527225 & 5013576814 & 6548689990 & 4555065871 \\ 0 & 4 & 0 & 2 & 1 & 2 & 3 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 2 & 3 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 3 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 47)y^3 + (-32s - 65038)y^2 + (569s + 347968)y + (-6358s - 6782218)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 89y^8 - 129922y^7 + 37300670y^6 - 2334788134y^5 + \\ & 82797719975y^4 - 1830224060950y^3 + 27558076586615y^2 - 26767357 \\ & 2155582y + 1515127960082636 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right)s, \quad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1553187443925056 = 2^6 \cdot 31^2 \cdot 59 \cdot 428024371$:

$$\begin{pmatrix} 6262852596472 & 4952689442050 & 3385043123645 & 5262592009628 & 4682207914831 & 2454240298106 & 3080031937494 & 2478168062243 & 4658895014553 & 3445007334746 \\ 0 & 62 & 31 & 26 & 31 & 22 & 32 & 42 & 31 & 27 \\ 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 3 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (s - 47)y^3 + (-32s - 65038)y^2 + (569s + 347968)y + (-6358s - 6782218)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 89y^8 - 129922y^7 + 37300670y^6 - 2334788134y^5 + 82797719975y^4 - 1830224060950y^3 + 27558076586615y^2 - 267673572155582y + 1515127960082636$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right)s, \quad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1114015603054132 = 2^2 \cdot 59 \cdot 16381 \cdot 288163427$:

$$\begin{pmatrix} 557007801527066 & 316934237403910 & 187216643674956 & 285774196366499 & 258780131924718 & 38327651433291 & 473207594079892 & 511997291929061 & 438155034157406 & 384728200116386 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 2 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3908y^3 + (-18s - 256495)y^2 + (1278s - 2337894)y + (-18144s - 84030696)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 + 7820y^8 - 528640y^7 + 11623970y^6 - 2163565780y^5 + 59637214915y^4 - 1129837238442y^3 + 131670166703868y^2 - 1292544292278672y + 19026570918295872$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right)s, \quad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $831195214191 = 3^6 \cdot 1140185479$:

$$\begin{pmatrix} 10261669311 & 1981050183 & 4044114609 & 2075129937 & 141688725 & 853100163 & 9381716019 & 2390399262 & 8743021250 & 9683745395 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 354 factors with signed exponents of absolute value up to 368331516325253843; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \ 4 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3908y^3 + (-18s - 256495)y^2 + (1278s - 2337894)y + (-18144s - 84030696)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 7820y^8 - 528640y^7 + 11623970y^6 - 2163565780y^5 \\ & + 59637214915y^4 - 1129837238442y^3 + 131670166703868y^2 - 12925 \\ & 44292278672y + 19026570918295872 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right)s, \quad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $15351132355983 = 3^7 \cdot 103 \cdot 1303 \cdot 52301$:

$$\begin{pmatrix} 189520152543 & 20575260729 & 58767465024 & 145265466492 & 165369457578 & 111040597788 & 114758162932 & 37286570917 & 67241036120 & 134200016196 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 & 3 & 0 & 1 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 & 3 & 0 & 2 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 3 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 360 factors with signed exponents of absolute value up to 493389506105579070; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + 3908y^3 + (-18s - 256495)y^2 + (1278s - 2337894)y + (-18144s - 84030696)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 + 7820y^8 - 528640y^7 + 11623970y^6 - 2163565780y^5 \\ & + 59637214915y^4 - 1129837238442y^3 + 131670166703868y^2 - 12925 \\ & 44292278672y + 19026570918295872 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ -1 \ 0 \ 1 \ 0 \ 0 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right)s, \quad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $859350175010928 = 2^4 \cdot 3^{10} \cdot 367 \cdot 1319 \cdot 1879$:

$$\begin{pmatrix} 49116951018 & 16467902424 & 14811877458 & 43459052580 & 37915200955 & 7014715698 & 24982371281 & 10672102490 & 47978491072 & 47515955827 \\ 0 & 18 & 0 & 9 & 0 & 6 & 6 & 14 & 2 & 15 \\ 0 & 0 & 6 & 3 & 5 & 0 & 2 & 2 & 4 & 4 \\ 0 & 0 & 0 & 27 & 1 & 6 & 2 & 6 & 2 & 25 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 2 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 355 factors with signed exponents of absolute value up to 71692846783669673; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (3 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 5451)y^3 + (55s + 197846)y^2 + (-751s - 3761924)y + (1488s + 51286436)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 10902y^8 + 406650y^7 + 58140293y^6 - 6044912354y^5 \\ & + 244590967708y^4 - 5158451024552y^3 + 60896872875187y^2 - 4671 \\ & 38484524188y + 2710840781434192 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right)s, \quad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1077847310102068 = 2^2 \cdot 11^2 \cdot 1334563 \cdot 1668679$:

$$\begin{pmatrix} 538923655051034 & 278237979419006 & 190485077547198 & 421419002106049 & 84552641985817 & 351988813244474 & 306920509858934 & 521180535255022 & 12921973035717 & 162488895613597 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 475 factors with signed exponents of absolute value up to 96091855382639210730; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (3 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 5451)y^3 + (55s + 197846)y^2 + (-751s - 3761924)y + (1488s + 51286436)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 10902y^8 + 406650y^7 + 58140293y^6 - 6044912354y^5 \\ & + 244590967708y^4 - 5158451024552y^3 + 60896872875187y^2 - 4671 \\ & 38484524188y + 2710840781434192 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right)s, \quad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $704056868913653 = 1249511 \cdot 563465923$:

$$\begin{pmatrix} 704056868913653 & 677929875813929 & 86723393268919 & 563709172198755 & 322507934507511 & 630525880570571 & 586832490503761 & 661861694017147 & 245146328672023 & 386453179794760 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 480 factors with signed exponents of absolute value up to 114222208111073801679; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (-s - 5451)y^3 + (55s + 197846)y^2 + (-751s - 3761924)y + (1488s + 51286436)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 - 10902y^8 + 406650y^7 + 58140293y^6 - 6044912354y^5 \\ & + 244590967708y^4 - 5158451024552y^3 + 60896872875187y^2 - 4671 \\ & 38484524188y + 2710840781434192 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \ 1 \ -1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right)s, \quad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $357105834361012 = 2^2 \cdot 11 \cdot 79^3 \cdot 16461257$:

$$\begin{pmatrix} 28609664666 & 25753292476 & 1995026400 & 4706669011 & 4624920681 & 585410048 & 12633787598 & 18175140166 & 9766273491 & 25609712223 \\ 0 & 79 & 0 & 1 & 49 & 65 & 29 & 14 & 0 & 21 \\ 0 & 0 & 158 & 127 & 78 & 69 & 89 & 98 & 139 & 35 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 476 factors with signed exponents of absolute value up to 53839552848120232929; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.